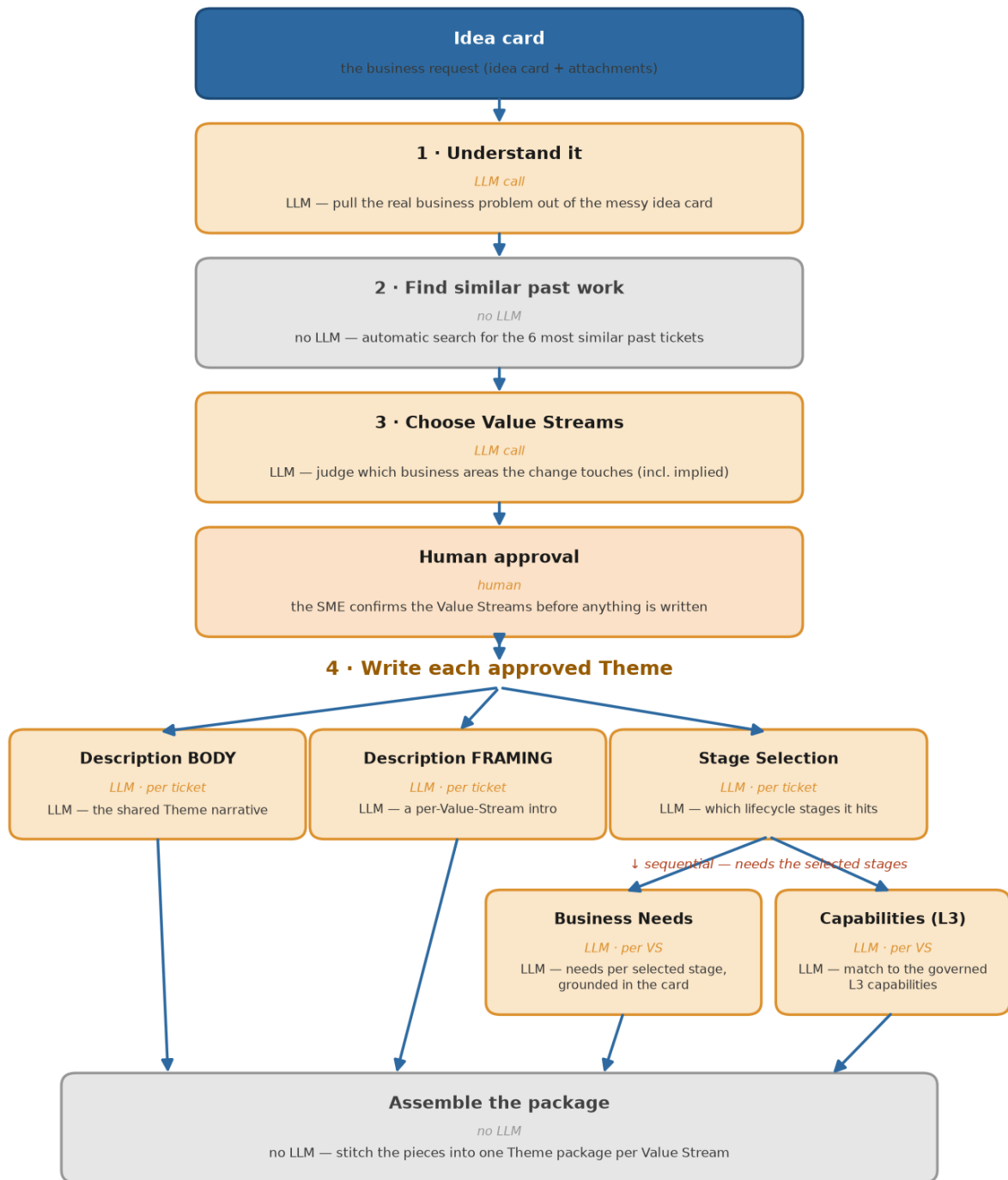


Generation flow — idea card to Theme packages

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Why parallel — and the one place it's sequential

- Description is independent of the stages, so it runs alongside everything and feeds the package directly.
- Stage Selection and the per-Value-Stream calls are batched and run in parallel where they can.
- The ONE sequential link: Business Needs and Capabilities need the SELECTED STAGES, so they wait for Stage Selection. Nothing else waits — wall-clock stays ~flat (≈15s) as Value Streams grow.

The steps

#	step	LLM ?	what it does	why it needs (or doesn't need) an LLM
	Idea card	—	The business request (idea card + attachments).	input
1	Understand it	LLM	Read the idea card and pull out the real business problem, capability and key terms.	The idea card is messy free text; a keyword scan can't extract meaning.
2	Find similar past work	no	Find the 6 most similar past tickets as precedent.	"Find similar" is a search — done automatically, no LLM.
3	Choose Value Streams	LLM	Decide which business areas (Value Streams) the change touches.	Needs judgment — especially the <i>implied</i> areas a keyword rule would miss.
	Human approval	human	The SME confirms the Value Streams.	Nothing is written until a person approves.
4	Write each approved Theme	LLM	For each approved Value Stream, write the stages , description , business needs and L2/L3 capabilities .	This is the writing work the tool is replacing.
	Assemble the package	no	Stitch the pieces into the final Theme package for review.	Mechanical assembly — no LLM.

Inside step 4 — how the Theme is written (parallel vs sequential)

Writing a Theme is not one call — it's a few, and most of them run **at the same time**. The calls fall into two bands:

Ticket-level — 3 calls, run in parallel (independent of each other):

- **Stage Selection** — which lifecycle stages the work hits, for *all* value streams in one batched call.
- **Description BODY** — the shared Theme narrative.
- **Description FRAMING** — a short per-value-stream intro, batched in one call.

These three don't depend on each other, so they fire together.

Per value stream — 2 calls each, run in parallel (and fanned out across all the value streams):

- **Business Needs** — the needs for each selected stage.
- **Capabilities (L3)** — the L3 capabilities the work exercises.

The one sequential dependency. Business Needs and Capabilities both need to know **which stages were selected**, so they **wait for Stage Selection** to finish — that is the only place the flow is forced to be sequential. Everything else overlaps.

What each LLM call sends and gets back

Each call is one focused request. Some run **once for the whole ticket**, some run **once for all value streams together (batched)**, and some run **once per value stream**. "Batched" is the key efficiency: one call covers every value stream instead of one call each.

call	how it runs	what we send in	what we get back
Understand (condense)	one call · per ticket	the idea-card text	a clean summary + the key business facts
Choose Value Streams	one call · per ticket	the idea-card text + all 50 Value Streams + the 6 similar past tickets	the chosen Value Streams, each with a reason
Stage Selection	one batched call · all Value Streams at once	the idea-card text + each Value Stream's candidate stages	the selected stages, grouped by Value Stream
Description BODY	one call · per ticket	the idea-card text	the shared Theme narrative
Description FRAMING	one batched call · all Value Streams at once	the idea-card text + the approved Value Streams	a short intro paragraph per Value Stream
Business Needs	one call · per Value Stream	the idea-card text + that Value Stream's selected stages	the Business Needs write-up for that Value Stream
Capabilities (L3)	one call · per Value Stream	the idea-card text + each selected stage's candidate L3 capabilities	the chosen L3 capabilities, per stage

Reading it: the two **batched** calls (Stage Selection, Description Framing) handle *all* value streams in a single request — that's why adding value streams doesn't multiply those calls. Only **Business Needs** and **Capabilities** run once per value stream, because each one writes value-stream- specific content. Every call is given only the idea-card text plus the governed options it must choose from — it never invents stages or capabilities, it selects from the approved lists.

Why there are several LLM calls (and why that's not brute force)

Each LLM call maps to **one decision a Business Architect would otherwise make manually** — understand the ticket, choose the areas, then write each Theme. For **N** approved Value Streams a request makes a small, fixed set of calls:

- **1** to understand the idea card,
- **1** to choose the Value Streams,
- a few to **write each approved Theme** (stages, description, needs and capabilities).

Everything else — finding similar past work, and assembling the final package — runs automatically with no LLM. So the model is used **only where the work is judgment or writing**; the mechanical parts deliberately stay code.

How many LLM calls per ticket

A ticket makes a small, predictable number of calls — **5 fixed, plus 2 per approved Value Stream**.

call	when it runs	how many
Understand (Condense)	once per ticket	1
Choose Value Streams	once per ticket	1
Stage Selection	once per ticket (batched — all Value Streams)	1
Description BODY	once per ticket	1
Description FRAMING	once per ticket (batched — all Value Streams)	1
Business Needs	once per approved Value Stream	N
Capabilities (L3)	once per approved Value Stream	N
Total		5 + 2N

approved Value Streams (N)	total LLM calls
1	7
3	11
10	25

The first **5 calls are fixed** no matter how many Value Streams are approved — the two batched calls (Stage Selection, Description Framing) cover all of them in one request. Only **Business Needs** and **Capabilities** add a call per Value Stream, because each writes Value-Stream-specific content.

Cost per ticket (GPT-5-mini)

GPT-5-mini pricing: **\$0.25 / 1M input tokens, \$2.00 / 1M output tokens**. For a ticket with **10 approved Value Streams**:

call	runs	input tokens	output tokens	cost
Condense	1	6,000	500	\$0.0025
Choose Value Streams	1	12,000	1,500	\$0.0060
Stage Selection	1 (batched)	7,602	1,273	\$0.0044
Description BODY + FRAMING	2	10,304	1,706	\$0.0060
Business Needs	10 (per VS)	55,200	15,670	\$0.0451
Capabilities (L3)	10 (per VS)	58,470	6,990	\$0.0286
Total		149,576	27,639	\$0.093

- **10 Value Streams: ~\$0.09 per ticket** — about **\$9 per 100 tickets, \$93 per 1,000**.
- **3 Value Streams: ~\$0.04 per ticket** — about **\$4 per 100 tickets, \$41 per 1,000**.

Only **Business Needs** and **Capabilities** scale with the number of Value Streams; the rest is fixed per ticket.